

Software Design Description 400

FPGA Image Processing and Display Unit (Tang Nano 9K)

References

- SDD000 - Development standards and architectural principles
- FPGA Budget - Resource utilization and timing analysis
- Pin Map - Complete pin allocation reference

Overview

Hardware Platform

Component	Specification
FPGA	Gowin GW1NR-9C (Tang Nano 9K)
LUTs	8640 total (4200 budgeted)
BRAM	26 blocks @ 2KB each (52KB total)
Clock	27 MHz system clock
Board	Tang Nano 9K development board

Development Toolchain

Tool	Version	Purpose
Yosys	Latest (open-source)	HDL synthesis
nextpnr-himbaechel	Latest	Place and route (Gowin family)
gowin_pack	Latest	Bitstream generation
openFPGALoader	Latest	FPGA programming

Build System: GNU Make (see [Makefile](./Makefile) in FPGA_Unit directory)

Language: Verilog (IEEE 1364-2005)

Functional Scope

The FPGA subsystem performs three primary functions:

1. Image Data Reception

- **Interface:** SPI slave receiving from ESP32-DevKit
- **Protocol:** SPI Mode 0 (CPOL=0, CPHA=0)
- **Data rate:** ~2 Mbps (limited by SPI clock)
- **Buffering:** 32KB dual-port BRAM for frame storage

2. Image Compression & Transmission

- **Algorithm:** 2D Walsh-Hadamard Transform (WHT) with sequency masking
- **Implementation:** Custom combinational `wht8.v` butterfly, no IP cores
- **Output:** Thresholded and quantized WHT coefficients returned to ESP32 via SPI (SOMI)
- **Compression ratio:** ~120:1 (76.8 KB -> ~500 B typical)
- **Detail:** See SDD401 — WHT Compression Pipeline

3. Local Display Management

- **Display:** 480x272 RGB TFT LCD
 - **Interface:** Parallel RGB565
 - **Features:** Test pattern generation, button-cycled modes
 - **Frame source:** BRAM buffer or test patterns
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Architecture

Module Hierarchy

```
top.v
|---- esp_interface.v      (SPI slave RX + MISO TX path)
|---- bram_image_buffer.v  (32KB dual-port buffer -- LCD display)
|---- lcd_controller.v     (RGB LCD timing generator)
|---- i2c_gpio_expander.v  (I2C slave -- stepper + buttons)
|---- gowin_rp11.v         (PLL, not used -- sys_clk = 27 MHz direct)
|---- block_sched          (blk_schd.v -- monolithic WHT pipeline FSM)
|   |---- wht2d.v          (wht8 wrapper, combinational 8-point 1D)
|   |   \---- wht8.v       (3-stage butterfly, zero latency)
|   |---- seq_msk.v        (combinational sequency mask, r+c > T)
|   \---- coeff.v          (Top-K coefficient extractor FSM)
```

Note: The planned modular decomposition in SDD401 (blk_read.v, wht_2d.v, coeff_out.v, somi.v, tx_fifo.v, psram_ctrl.v) was not implemented as described. See SDD401 for details.

Data Flow Pipeline

```
ESP32-DevKit (SPI Master)
| SPI (MOSI, SCLK, CS_n, MISO)
v
esp_interface.v          -- RX: byte stream in, MISO: coefficients out
|----> BSRAM write        ----> lcd_controller.v -> RGB LCD (480x272)
\----> PSRAM write (cur_base)

[CS rising edge -> cs_end -> block_sched.start]

block_sched.v (blk_schd.v)
-- LOAD: reads cur_base + prev_base word pairs from PSRAM
-- Diffs each pair, fills 64-entry pixel register array
-- WHT: 16 passes (8 row + 8 col) via wht2d.v/wht8.v (combinational)
-- seq_msk.v: zeros coefficients where r+c > 6 (combinational)
-- WAIT: coeff.v FSM scans 64 positions, Top-5 by magnitude
-- EMIT: 2 header bytes + 3 bytes per coefficient -> tx_buf in top.v
-- NEXT: advance block counters, repeat for all 1200 blocks
-- frame_done -> top.v swaps cur_base/prev_base

top.v tx_buf (512-byte BRAM)
-- sched_tx_valid writes coefficients in as they arrive from block_sched
-- esp_tx_next advances tx_rd_ptr; tx_rd_data feeds esp_interface.v tx_data
-- ESP32 reads back coefficients via SPI (MISO path in esp_interface.v)
```

See SDD401 for full WHT pipeline design and implementation notes.

Module Descriptions

top.v - System Integration

Purpose: Top-level module connecting all subsystems

Key Features: - Power-on reset generation (255-cycle delay) - SPI-to-BRAM data path management - Frame synchronization via CS edge detection - Debug LED indicators for system state

Interfaces: - **Input:** 27MHz system clock, SPI slave signals, button - **Output:** RGB LCD signals, debug LEDs

Critical Logic: - CS falling edge -> reset write address, begin frame reception - CS rising edge -> mark frame complete, set `frame_ready` flag - Continuous streaming: byte valid -> write to BRAM, increment address

esp_interface.v - SPI Slave Controller

Purpose: Full-duplex SPI slave – RX path ingests image bytes, MISO TX path returns compressed coefficients.

Implementation Details: - **Clock domain crossing:** 3-stage synchronizer on SCLK and CS_n; 2-stage on MOSI - **Edge detection:** Rising (sample) and falling (shift) detection for SCLK - **RX shift register:** 8-bit serial-to-parallel, byte_ready strobe - **TX shift register (tx_sr):** Preloaded from tx_data on byte boundary and CS assert - **MISO output:** tx_sr[7] gated by cs_active

SPI Protocol: - Mode 0: Sample on rising edge, shift on falling edge - MSB-first bit ordering - Active-low chip select

Known Bug (Bug 1): Inter-byte MISO shift. After tx_sr is reloaded on byte_done_d, the immediately following falling SCLK edge shifts tx_sr left by 1 before the master samples the new MSB. Every byte[N>=1] received by ESP32 is the correct byte shifted left by 1. Confirmed by esp_interface_tx_tb (22 failures). See UTR501.

Current Status: [DONE] Implemented – Bug 1 open

bram_image_buffer.v - Image Frame Buffer

Purpose: Dual-port RAM for concurrent write (SPI) and read (LCD/FFT)

Implementation: - **Type:** Inferred single-clock dual-port RAM - **Size:** 32KB (15-bit address space) - **Synthesis:** Gowin BSRAM primitives (automatically inferred)

Ports: - **Write:** SPI data stream (byte-wide) - **Read:** LCD controller (byte-wide, grayscale)

Note: Dedicated to LCD display path only. Frame storage for WHT processing uses embedded PSRAM (see SDD401).

Current Status: [DONE] Functional

lcd_controller.v - RGB Display Driver

Purpose: Generate RGB LCD timing and display content

LCD Specification: - **Resolution:** 480x272 pixels - **Interface:** RGB565 (5-bit R, 6-bit G, 5-bit B) - **Timing:** HSYNC, VSYNC, DE (data enable), pixel clock

Features: - **Test patterns:** 8 modes (solid colors, bars, gradients, checkerboard, BRAM data) - **Button control:** Debounced button cycles through patterns - **BRAM display mode:** Pattern 7 displays frame buffer content

Timing Parameters:

Horizontal: 480 active + 2 front + 41 sync + 2 back = 525 total

Vertical: 272 active + 2 front + 10 sync + 2 back = 286 total

Pixel clock: 13.5 MHz (27 MHz / 2)

Current Status: [DONE] Fully functional with test patterns

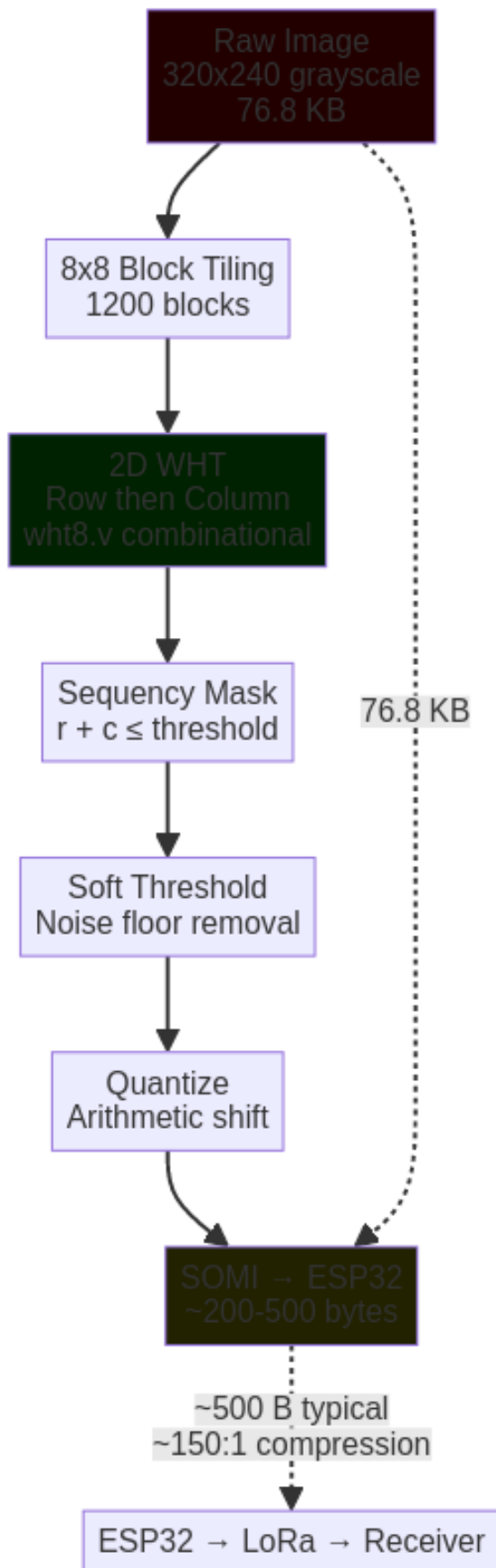
WHT Compression Module

See **SDD401 — WHT Compression Pipeline** for full design.

Summary

The compression pipeline uses a 2D Walsh-Hadamard Transform on 8x8 pixel blocks — custom RTL, no vendor IP cores. The `wht8.v` module is a combinational 3-stage butterfly (zero multipliers, additions/subtractions only) that maps directly from the branchless FWHT class in `sim/wht_core.py`.

Compression Pipeline:



Reconstruction (Receiver Side)

Performed on ESP32-S3: 1. Receive thresholded WHT coefficients via LoRa 2. Dequantize and place into 8x8 coefficient blocks 3. Inverse 2D WHT (same butterfly, divide by 64) 4. Gaussian smoothing on reconstructed diff 5. Add diff to reference frame

Rationale: Inverse WHT is cheap (same add/sub butterfly). All heavy lifting stays on the FPGA; the receiver just reverses the transform.

Resource Utilization

Actual Utilization (Synthesized)

Module	LUTs	BRAM	Status
LCD RGB Controller	~800	0	[X]
SPI Slave (RX+TX)	~250	0	[X]
I2C GPIO Expander	~400	0	[X]
Control logic	~300	0	[X]
BRAM buffer (LCD)	(inferred)	16	[X]
tx_buf (top.v)	(inferred)	2	[X]
block_sched	~2000	0	[X]
wht8 (via wht2d)	~200	0	[X]
seq_msk	~50	0	[X]
coeff	~300	0	[X]
Estimated Total	~4300	18	[X]

Margin: ~4340 LUTs remaining (50% used), 8 BRAM blocks unused. PSRAM is embedded (no LUT/BRAM cost). Synthesis report in build/ is authoritative.

Timing Budget

Image Processing Pipeline

ESP32-CAM -> ESP32-DevKit: Forward image
\--- SPI transfer: ~38 ms (2 Mbps, 76.8 KB frame)

FPGA: Receive + Process
\--- SPI reception: ~38 ms (concurrent with above)
\--- 2D WHT all 1200 blocks: ~19 ms (see SDD401 timing)
\--- Coefficient readback via SOMI: ~1 ms
\--- Total FPGA processing latency: ~20 ms

FPGA processing completes before next frame arrives - no backpressure.

Interfaces

SPI Slave (from ESP32-DevKit)

Signal	Direction	Pin	Description
esp_mosi	Input	49	Master Out Slave In (data)
esp_miso	Output	77	Master In Slave Out (unused)
esp_sclk	Input	76	SPI clock from master
esp_cs_n	Input	48	Chip select (active low)

Protocol: SPI Mode 0, byte-oriented streaming

SOMI — SPI Return Path (SDD401)

Signal	Direction	Pin	Description
esp_miso	Output	76	Compressed coefficients (reuses existing SPI MISO pin)
tx_ready	Output	TBD	High when coefficient data is ready for readback

Protocol: ESP32 initiates SPI read (clocks in bytes on MISO) after `tx_ready` asserts. See SDD401 for full SOMI design.

RGB LCD Interface

Signal	Direction	Pins	Description
lcd_clk	Output	35	Pixel clock (13.5 MHz)
lcd_de	Output	33	Data enable
lcd_hsync	Output	40	Horizontal sync
lcd_vsync	Output	34	Vertical sync
lcd_r[4:0]	Output	71, 73, 75, 74, 72	Red channel
lcd_g[5:0]	Output	69, 70, 57, 68, 56, 55	Green channel
lcd_b[4:0]	Output	54, 53, 51, 42, 41	Blue channel

Format: RGB565 (16-bit color depth)

Build System

Makefile Targets

```

make           # Synthesize, place & route, pack bitstream
make program   # Build and flash to FPGA
make flash     # Flash existing bitstream (skip rebuild)
make clean     # Remove build artifacts

```

Build Flow

1. **Synthesis:** Yosys converts Verilog -> JSON netlist
2. **Place & Route:** nextpnr-himbaechel maps to FPGA fabric
3. **Pack:** gowin_pack generates .fs bitstream
4. **Program:** openFPGALoader flashes via USB-JTAG

Build time: ~30 seconds on typical development machine

Development Status

Completed [DONE]

- ☒ SPI slave RX path with 3-stage synchronizer, byte_ready strobe
- ☒ SPI MISO TX path (tx_sr, preload from tx_data, byte_done_d reload)
- ☒ BRAM image buffer for LCD (dual-port, 32KB)
- ☒ PSRAM dual-write during SPI ingest (cur_base/prev_base ping-pong)
- ☒ RGB LCD controller with test patterns
- ☒ Top-level integration and frame synchronization
- ☒ Button debouncing and pattern cycling
- ☒ I2C GPIO expander (stepper motor + joystick buttons)
- ☒ Debug LED indicators
- ☒ Build system and toolchain setup
- ☒ wht8.v – 8-point combinational WHT butterfly (wht8_tb PASS)
- ☒ wht2d.v – wht8 bus wrapper for blk_schd
- ☒ seq_msk.v – combinational sequency mask ($r+c > 6 \rightarrow \text{zero}$)
- ☒ coeff.v – Top-K extractor FSM (thresh=400, q_shift=3, TOP_K=5, coeff_tb PASS)
- ☒ blk_schd.v – monolithic block scheduler FSM (IDLE/LOAD/WHT/WAIT/EMIT/NEXT)
- ☒ top.v tx_buf (512-byte BRAM, registered read, tx_rd_ptr, tx_len)
- ☒ Frame diff (FPGA-side, cur_base vs prev_base)

Open Bugs

- ☐ Bug 1: MISO inter-byte shift in esp_interface.v – 22 sim failures (UTR501 TB003)
- ☐ Bug 2: blk_schd_tb PSRAM mock deadlock – testbench hangs (UTR501 TB004)

Future Work [TODO]

- ☐ Fix Bug 1: MISO TX pipeline skew
 - ☐ Fix Bug 2: rewrite blk_schd_tb PSRAM mock with explicit cycle counter
 - ☐ Power management (watchdog, sleep modes)
 - ☐ Dynamic compression ratio adjustment (runtime SEQ_THRESH / BLOCK_THRESH)
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Testing Strategy

Unit Testing

- **SPI Interface:** Verified with ESP32 sending test patterns
- **LCD Controller:** All 8 test patterns validated on physical display
- **BRAM:** Read/write verification via pattern mode 7

Integration Testing

- **End-to-end:** ESP32 $\rightarrow$ SPI $\rightarrow$ FPGA $\rightarrow$ LCD (functional, camera stream visible)
- **WHT Pipeline:** wht8_tb PASS, coeff_tb PASS; blk_schd_tb blocked by testbench bug (UTR501)

Debug Capabilities

- **LEDs:** 6 debug LEDs indicate system state (receiving, frame ready, CS, etc.)
 - **Test patterns:** Button-cycled patterns verify LCD timing
 - **Serial monitor:** UART debug output (to be added)
-

Known Issues & Limitations

1. **Bug 1 – MISO inter-byte shift:** `esp_interface.v` `tx_sr` is reloaded on `byte_done_d`, but the next falling SCLK edge (the between-bytes transition) immediately shifts it left by 1. Every `byte[N>=1]` received by the ESP32 equals the correct byte shifted left by 1 (MSB appears as 0, bit 0 of the correct byte is lost). Confirmed by sim, 22 failures. Fix: gate the shift on `bit_count != 0` (skip the first falling edge after reload). See UTR501 TB003.
 2. **Bug 2 – `blk_schd_tb` PSRAM mock deadlock:** The testbench PSRAM model clears `psram_busy` in the same cycle that `blk_schd` can fire a second read (when `psram_busy` reads as stale 0). `blk_schd` clears `psram_pending` before the mock delivers data, leaving `psram_busy` stuck at 1 permanently. Simulation hangs. Fix: rewrite mock with an explicit N-cycle countdown rather than edge detection. See UTR501 TB004.
 3. **`top.v` `tx_buf` registered-read lag:** BRAM read has 1-cycle latency. When `byte_done_d` fires to reload `tx_sr`, `tx_rd_data` still holds the byte from the previous `tx_rd_ptr` value. `tx_rd_ptr` must advance one cycle before the reload fires, or `tx_sr` preload must be delayed by one additional cycle.
 4. **ESP32 heap corruption (`cam_spi` buffer leak):** 76800-byte DMA buffers are leaked on RX checksum failure. Over time this corrupts the heap region holding `ctx.mutex`, causing an `xSemaphoreGive` assert crash at `fpga_spi.c:192`. Fix: call `safe_free` in `cam_spi` `rx_reset` path before `realloc`.
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References

Vendor Documentation

- Gowin GW1NR-9 FPGA Family Datasheet
- Tang Nano 9K Schematic & Pinout
- GW1NR-9C Embedded PSRAM Usage — QSPI interface to onboard 64 Mbit PSRAM

Open-Source Toolchain

- Yosys Open Synthesis Suite
- nextpnr (Himbaechel for Gowin)
- openFPGALoader

Related SDDs

- SDD401 — WHT Compression Pipeline
- SDD003 — LoRa Link Budget
- SDD004 — LoRa Bench Test & Debug

Standards

- IEEE 1364-2005 (Verilog HDL)
- SPI Protocol (Motorola/IEEE)
- RGB LCD Timing Specifications (VESA standards)